# Automated Nucleic Acid Extractor

# User Manual

## Product information

Product: Automated Nucleic Acid Extractor

Specification: Dual-channel Extractor

## Manufacturer

## Version information

This manual may be updated without further notice.

Version number: 1.2

Release date: 2023.12

## International Symbols Description

| Symbol | Description | Symbol | Description |
|---|---|---|---|
| | **Keep away from open flame** | | **Keep dry** |
| | **This side up** | | **Handle with care** |
| | **Fragile** | | **Keep away from sunlight** |

Table 1. Symbol Description

# Technical Support and Warranty

## Maintenance

This product is guaranteed for one year. Please keep the packing box for transportation, storage, or for use when product needs to be shipped for maintenance.

We are responsible for the safety, reliability and performance of this product under the following conditions:

- The instrument is assembled, upgraded, or maintained by an authorized party.
- Relevant electrical equipment complies with national standard.
- This product is used in accordance with this instruction manual.

If the product is in the following circumstances, which is not covered by the warranty, the consumer may need to pay the full amount for maintenance costs.

- The product is damaged due to the following reasons:
  - ◆ Improper operation
  - ◆ Improper connection with other devices
  - ◆ Accidents
- Modification of products without authorization
- Unrecognized product serial number

## Technical Support

If technical support is needed, please contact your local distributor.

Please identify the following information:

- Type of malfunction or damage
- Product model and serial number

# Safety Notice

### Warning messages

The terms "danger", "warning", "caution", and "attention" used in the manual indicate hazards and remind users to pay attention to the process of operation according to the degree of hazards. A hazard is a source of potential harm or adverse effects to the environment or user.

| | | |
|---|---|---|
| | **Attention** | Provide users with useful information and tips to ensure maximum performance of this product. |
| | **Caution** | Refers to a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. |
| | **Warning** | Refers to a hazardous situation which, if not avoided, could result in death or serious injury. |
| | **Danger** | Refers to an immediate hazard which, if not avoided, will result in death or serious injury. |

### Safety Information

| | | |
|---|---|---|
| | **Warning** | 1. Use a reliable, grounded power outlet.<br>2. Without the manufacturer's permission or written authorization, the user should not disassemble, modify, or use various accessories that do not comply with the manufacturer's regulations.<br>3. Before cleaning or disinfecting the instrument, the instrument must be turned off and the power cord must be unplugged.<br>4. Do not disassemble the instrument to prevent electric shock. |
| | **Caution** | 1. The instrument must be used with special consumables.<br>2. Do not interfere the operation of the instrument or move the stationary parts of the instrument during operation. |
| | **Attention** | Please read the instructions carefully before use. |

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# Chapter 1 System Introduction

This manual provides general information and operating instructions of the nucleic acid extractor.

## 1.1 Introduction

The Automated Nucleic Acid Extractor is an automated equipment used for extraction of sample nucleic acid for PCR detection.

## 1.2 Product features

Fast extraction: 2 samples can be processed at the same time, in as little as 8 minutes.

Safe and reliable: The instrument runs automatically without manual operations. Reagent and magnetic cover are clearly marked for easy identification to ensure proper usage and to avoid leakage, ensuring high reliability in the test.

Contamination control: Small and closed extraction chamber design with built-in ultraviolet light can effectively remove potential nucleic acid contamination.

User-friendly: Built-in operation program with one-button start.

# Chapter 2 Instrument Introduction

## 2.1 Components

The Automated Nucleic Acid Extractor is 2-channel fully automatic nucleic acid extractor, which needs to be equipped with special consumables (reagent kit and magnetic cover) for operation.

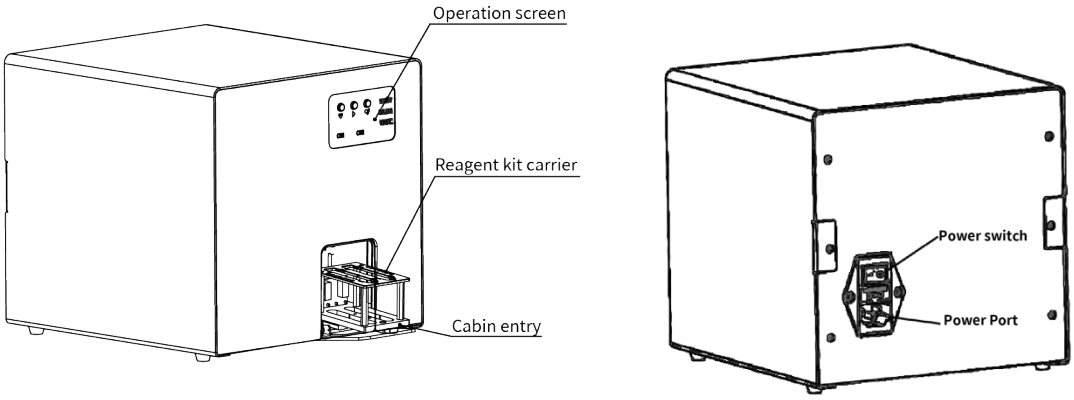


Fig1. Appearance of the Automated Nucleic Acid Extractor

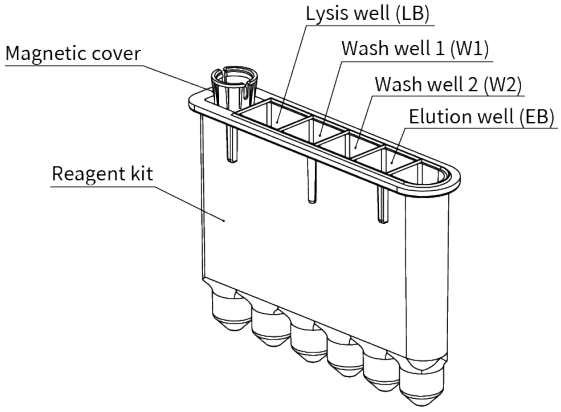


Fig2. Diagram of dedicated consumables

The magnetic cover is placed in the position as shown on the kit. The instrument can automatically grasp it before the extraction starts and automatically return it to the original position after the extraction is completed. The reagents are preset in the kit and the sample is added to the lysis well before extraction.

## 2.2 Specifications

| | |
|---|---|
| Methodology: | Magnetic bead |
| **Specifications** | |
| Dimensions: | 226*207*205 mm |
| Weight: | 3.7 Kg |
| Power supply: | AC 110V/220V 50/60HZ |
| Storage: | Keep away from moisture |
| Difference between Channels: | CV ≤ 5% |

| **Operation environment** | |
|---|---|
| Temperature: | 10℃ - 40℃ |
| Relative moisture: | 10% - 90%RH |
| Altitude: | 2000 meters |
| Noise: | <70dB |
| Magnetic bead recovery: | >92% |
| Extraction Time: | <8min |
| Oscillation: | 3HZ |
| Contamination control: | UV sterilization |

## 2.3 Operation screen

Please refer to the figure below for the key functions and indicators on the operation screen.

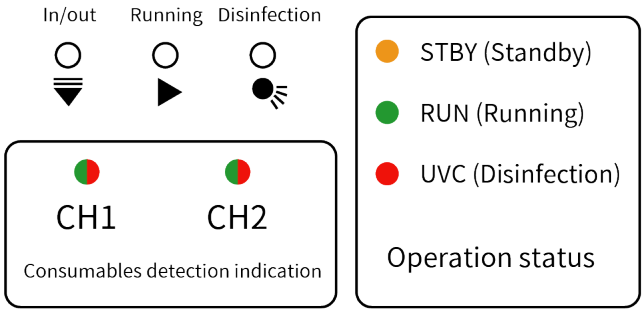


Fig3. Operation screen

### 2.3.1 Indicator lights

| **Consumables detection indication** |
|---|
| The indicator light has three states: off, red, and green. |
| Off: Unloaded |
| Red: Kit detected |
| Green:Magnetic cover detected |

| **Operating status indication** |
|---|
| Standby: Yellow light flashes |
| Running: Green light on |
| Disinfection: Red light on |

### 2.3.2 Key function description

**In/out:** Control the reagent kit carrier between the out-of-cabin and in-cabin state. Controlled exit is only valid in the standby state of the instrument and is invalid in the running or disinfection state.

**Running:** Control the operation of the extractor. Press to start the extraction program. It is only valid when the reagent kit is in the cabin and the instrument is in the standby state. It is invalid during the running or disinfection state of the instrument. Long press to forcibly stop the operation.

**Disinfection:** Turn the UV sterilization on or off. Press to turn on the UV. It is only valid when the reagent kit is in the cabin and the instrument is in the standby state. It is invalid in the running state. Press it again to turn off the UV manually, or it will automatically turn off after 15 minutes.

# Chapter 3 Getting started

## 3.1 Packaging contents

When you receive the Automated Nucleic Acid Extractor, please check that it includes the following items.

| No. | Item | Quantity | Unit |
|---|---|---|---|
| 1 | Automated Nucleic Acid Extractor | 1 | Unit |
| 2 | Power cord | 1 | Piece |
| 3 | User manual | 1 | Booklet |
| 4 | Extraction kit holder | 1 | PCS |
| 5 | PCR rack | 1 | PCS |
| 6 | Certificate | 1 | Sheet |
| 7 | Warranty card | 1 | Sheet |
| 8 | List of product contents | 1 | Sheet |

## 3.2 Operating environment

1. Operating temperature: 20℃- 40℃

2. Relative humidity: 10% - 90%

3. This instrument should be used and kept in an indoor environment. Keep the instrument on a flat, and stable surface.

4. Do not place the instrument in a humid or dusty area. The room should be well ventilated and free from corrosive gas or strong magnetic field interference.

## 3.3 Operating environment

For the first time user, please follow the instructions. Connect the power supply according to the instructions and turn on the power switch. The instrument will immediately reset to its reference point and enter the standby state.

1. Prepare Consumables

Take the new reagent kit, shake it well, and place it in the reagent kit carrier. Then, tear off the sealing film, add 200μL of sample to the LB well, and place the magnetic cover in its well as shown in Figure 2.

2. Place Consumables

Press the key and the carrier will extend out. Then, place the reagent kit into the carrier and confirm

that it is properly in place (the 4 supports on the side of the kit will be aligned in the carrier), and note that the magnetic cover should be in the indicated position.

After the kit is placed, the consumable indicator light on the operation screen will change from off to red.

3. Extraction:

Press the enter/exit key and wait for the carrier to retreat into the cabin. Then, press the run key.

The extraction program will start to run, and the panel status indicator will turn from standby to running. Within 5 seconds, the consumables indicator turns from red to green, indicating that the magnetic cover has been successfully picked up and that the extraction process has begun. After about 8 minutes, the buzzer will sound, indicating that the extraction was completed. The panel status indicator will change from running to standby.

4. Eject consumables:

Press the enter/exit key and wait for the carrier to extend out. Then, take out the reagent kit and place it on the external carrier. Press the enter/exit key again to retract the carrier. Use a pipette to draw the liquid in the EB well for PCR testing.

5. Disinfection

Press the disinfection button while in the standby state. The UV lamp in the instrument will be turned on and it will automatically turn off after 15 minutes.

6. Turning off the instrument

After sterilization, directly turn off the power of the instrument with the switch.

# Chapter 4 Cleaning and Maintenance

The instrument is a closed structure. If there is no special situation, wiping with alcohol will be sufficient.

**Surface cleaning:** Wipe with a clean, dust-free cloth and 75% alcohol.

**Internal cleaning:** Turn on the ultraviolet lamp for disinfection or wipe it with 75% alcohol after extending the reagent kit carrier.

| | | |
|---|---|---|
| | **Danger** | Switch off and unplug the instrument before cleaning. |
| | **Caution** | Do not disassemble the instrument without permission, it may cause damage to the instrument |
| | **Caution** | If the instrument overturns during the operation, causing the reagents to flow out or the consumables to drop, please contact technical support. |

# Chapter 5 Troubleshooting

| No. | Problem | Possible causes | Solution |
|---|---|---|---|
| 1 | After the instrument starts running, the buzzer sounds and the instrument stops running. | The magnetic cover is not in the kit. | Place magnetic cover in the kit as required. |
| 2 | The reagent kit shakes severely when the instrument is running | The magnetic cover is deformed and bent, or the kit is not placed in the proper position. | Stop the process and check the magnetic cover and kit position. |
| 3 | No response after pressing the Exit or Run button. | Instrument may be in the disinfection state. | Check and turn off disinfection mode manually. |
| 4 | The instrument cannot operate, and the green light of the consumable detection indicator is on. The buzzer sounds after pressing the run button. | There is a magnetic cover left on the magnetic cover holder. | Load an empty reagent box, then press and hold the run button to perform the demagnetization action. Then, remove the leftover magnetic cover. |

If there are other abnormalities, please contact technical support.